CENG381 Stochastic Processes

Random Mapping Representation — Practice 1

Q1 (30 points). Consider the random walk on the 3-cycle C_3 with states $X = \{0, 1, 2\}$ (integers mod 3). The transition probability is:

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P(j, k) = 1/2   if k = j+1 mod 3
P(j, k) = 1/2   if k = j−1 mod 3
P(j, k) = 0     otherwise
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a) Write the 3×3 matrix P explicitly.

b) Define the random mapping $f(j, Z) = (j + Z) \bmod 3$, where Z is uniform on $\Lambda = \{-1, +1\}$. Verify that this is a valid representation by checking $P\{ f(j, Z) = k \} = P(j, k)$ for every j and k.

c) Starting from $X_0 = 0$, let $X_n = f(X_{n-1}, Z_n)$ with $Z_1, Z_2, \ldots$ i.i.d. uniform on $\{-1, +1\}$. Compute $P(X_1 = 1)$ and $P(X_2 = 0)$.

Q2 (35 points). Consider the random walk on the 4-cycle C_4 with states $X = \{0, 1, 2, 3\}$ (integers mod 4) and transition probabilities $P(j, k) = 1/2$ if $k = j \pm 1 \bmod 4$, and 0 otherwise.

a) Give a random mapping representation: define $f: X \times \Lambda \to X$ and a noise distribution for Z, and verify $P\{ f(j, Z) = k \} = P(j, k)$.

b) Starting from $X_0 = 0$ and using the representation $X_n = (X_0 + Z_1 + Z_2 + \ldots + Z_n) \bmod 4$, write a closed-form expression for X_n in terms of the partial sums $S_n = Z_1 + \ldots + Z_n$.

c) Suppose two copies of the chain are driven by the same noise: $X_0 = 0$ and $Y_0 = 2$, with $X_n = S_n \bmod 4$ and $Y_n = (2 + S_n) \bmod 4$. Show that $X_n - Y_n = -2 \bmod 4$ for all $n \geq 0$. What does this imply about coupling and convergence for this chain?

Q3 (35 points). Consider the Markov chain on states {0, 1, 2} with transition matrix:

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       0      1      2
0  [ 1/3    1/3    1/3 ]
1  [ 1/2    1/2     0  ]
2  [  0     1/2    1/2 ]
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a) Construct a random mapping representation using $Z \sim \text{Uniform}[0,1]$ (continuous). For each state x, define $f(x, z)$ by partitioning [0,1] into intervals corresponding to the outgoing transitions.

b) Is the chain irreducible? Is it aperiodic? Find the stationary distribution π.

c) Starting from $X_0 = 0$, compute the two-step distribution $\mu_2 = \mu_0 \cdot P^2$ by first finding $\mu_1 = \mu_0 \cdot P$, then $\mu_2 = \mu_1 \cdot P$. Is the chain already close to π after 2 steps?